# Op-amps, differential amps, power amps

Answers and worked solutions · Day 21 · Fri 13 Nov · pack E13

## 5. Practice

### From your course files

**WS10 Q1 (course question).**

> See Worked Example 1. $A_d = 120$, $A_c = 20$, CMRR $= 6$, **CMRR = 15.56 dB**.

**WS10 Q2 (course question).**

> See Worked Example 2. $V_d = 60$ mV, $V_c = 170$ mV.
>
> a. CMRR $= 200$: $V_o = 360\ \text{V}(1 + 0.01417) =$ **365.1 V**.
>
> b. CMRR $= 10^5$: $V_o = 360\ \text{V}(1 + 2.83 \times 10^{-5}) =$ **360.01 V**.
>
> If the inputs are read as µV (Boylestad's original), the answers are 365.1 mV and 360.01 mV. Mention that a real op-amp would saturate at its rails for the mV version.

**WS10 Q3 (course question).**

> READ THE FIGURE: $V_1$ goes straight into the **+** input (pin 5). A 36 kΩ resistor runs from the − input (pin 4) to ground, and 750 kΩ runs from the output (pin 10) back to −. Supplies are ±9 V (741). So this is a **non-inverting amplifier** with $R_1 = 36$ kΩ and $R_f = 750$ kΩ.
>
> $$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{750}{36} = 21.83$$
>
> $$V_o = 21.83 \times 150\ \text{mV} = \mathbf{3.275\ V\ rms}\ (\text{in phase with } V_1)$$
>
> Rail check: the peak is $3.275\sqrt{2} = 4.63$ V, which is inside ±9 V, so there is no clipping.
>
> Trap: $R_f/R_1 = 20.8$ alone would give 3.125 V. The "1 +" is worth the mark.

**WS10 Q4 (course question).**

> READ THE FIGURE: $V_1$ through 47 kΩ and $V_2$ through 12 kΩ both go to the − input (pin 4). There is 470 kΩ feedback, and + (pin 5) is grounded. Supplies are ±15 V. This is an **inverting summing amplifier**.

$$V_o = -\left(\frac{470}{47}V_1 + \frac{470}{12}V_2\right) = -\left(10(40\text{ mV}) + 39.17(20\text{ mV})\right)$$

$$V_o = -(400 + 783.3)\text{ mV} = \mathbf{-1.183\text{ V rms}}$$

So the output is 1.183 V rms, 180° out of phase with the (in-phase) inputs. The peak is 1.67 V, well inside ±15 V.

**WS10 Q5 (course question).**

READ THE FIGURE:

- **Stage 1** is an inverting amplifier: $V_1 = 12$ mV through 33 kΩ to −, with 330 kΩ feedback and + grounded.
- **Stage 2** is an inverting summer. Stage 1's output goes through 47 kΩ, and $V_2 = 18$ mV goes through another 47 kΩ. Both go to the − input, with 470 kΩ feedback and + grounded.

Stage 1:

$$V_{o1} = -\frac{330}{33}(12\text{ mV}) = -10(12\text{ mV}) = -120\text{ mV}$$

Stage 2:

$$V_o = -\left(\frac{470}{47}V_{o1} + \frac{470}{47}V_2\right) = -10\left(-120 + 18\right)\text{ mV}$$

$$V_o = -(-1200 + 180)\text{ mV} = \mathbf{+1.02\text{ V}}$$

The two inversions cancel on the $V_1$ path (+100 overall). $V_2$ is inverted once (−10). This assumes $V_1$ and $V_2$ are in phase (or are dc values), as the figure implies.

**WS10 Q6 (course question).**

**From Fig. 1 (page 3):** $V_{CC} = 18$ V, $R_B = 1.2$ kΩ, $R_C = 16$ Ω, $\beta = 40$, fixed bias.

DC bias:

$$I_B = \frac{18 - 0.7}{1.2\text{ k}\Omega} = 14.42\text{ mA}, \quad I_{CQ} = 40(14.42\text{ mA}) = 576.7\text{ mA},$$

$$V_{CEQ} = 18 - (0.5767)(16) = 8.77\text{ V (active region, close to centred)}.$$

Input power:

$$P_i(\text{dc}) = V_{CC}I_{CQ} = 18(0.5767) = \mathbf{10.38\text{ W}}.$$

Output power, with $I_b = 5$ mA rms:

$I_c(\text{rms}) = \beta I_b = 40(5\text{ mA}) = 200\text{ mA}, \qquad P_o(\text{ac}) = I_c^2(\text{rms})R_C = (0.2)^2(16) = \mathbf{0.64\text{ W}}.$

Check for clipping: $I_c(p) = 0.2\sqrt{2} = 283$ mA. This is below both $I_{CQ} = 577$ mA and $V_{CC}/R_C - I_{CQ} = 548$ mA, so the output is undistorted. Efficiency $= 0.64/10.38 = 6.2\%$, well under the 25% limit.

**WS10 Q7 (course question).**

Same circuit (values as in Q6) with $R_B = 1.5$ kΩ:

$$I_B = \frac{17.3}{1.5\text{ k}\Omega} = 11.53\text{ mA}, \quad I_{CQ} = 40(11.53) = 461.3\text{ mA}, \quad V_{CEQ} = 18 - 7.38 = 10.62\text{ V}.$$

$$P_i(\text{dc}) = 18(0.4613) = \mathbf{8.30\text{ W}}.$$

**WS10 Q8 (course question).**

With $R_B = 1.5$ kΩ: $I_{CQ} = 461.3$ mA. The saturation current is $V_{CC}/R_C = 18/16 = 1.125$ A.

The largest undistorted collector-current peak is the smaller of:

- the room down to cut-off, $I_{CQ} = 461.3$ mA, and
- the room up to saturation, $1.125 - 0.4613 = 663.7$ mA.

So $I_c(p) = 461.3$ mA (it clips at cut-off first, because the Q-point sits above the centre of the load line in $V_{CE}$).

$$P_o(\text{max}) = \frac{I_c^2(p)}{2}R_C = \frac{(0.4613)^2}{2}(16) = \mathbf{1.70\text{ W}}.$$

Efficiency at this output: $1.70/8.30 = 20.5\%$. It is below 25% because the Q-point is not centred. For comparison, a centred Q-point would give $V_{CC}^2/(8R_C) = 2.53$ W.

**WS10 Q9 (course question).**

Full working: Worked Example 4.

$R_L = 16 \parallel 16 \parallel 16 \parallel 16 = 4\ \Omega$. $a = \sqrt{8000/4} = \sqrt{2000} = \mathbf{44.72 : 1}$ (step-down).

**WS10 Q10 (course question).**

See Worked Example 3. (a) $P_i(\text{max}) = \mathbf{49.74\text{ W}}$; (b) $P_o(\text{max}) = \mathbf{39.06\text{ W}}$; (c) $\eta_{\text{max}} = \mathbf{78.54\%}$.

## New questions

**N1. [3 marks]**

a. CMRR $= 10^{80/20} = 10^4$, so $A_c = 5000/10^4 =$ **0.5**.

b. $V_d = 0.5$ mV, $V_c = 1.0$ mV.
$V_o = A_d V_d + A_c V_c = 5000(0.5\ \text{mV}) + 0.5(1.0\ \text{mV}) = 2.5\ \text{V} + 0.5\ \text{mV} =$ **2.5005 V**.

Check with the bracket form: $2.5(1 + 1.0/(10^4 \times 0.5)) = 2.5005$ V.

**N2. [5 marks]**

a. Stage 1: $V_{o1} = -(150/10)(0.2) =$ **−3.0 V**. Stage 2:
$V_o = (1 + 22/10)(-3.0) = 3.2(-3.0) =$ **−9.6 V** (fine with ±15 V rails).

b. $V_o = -\left(\frac{100}{20}(0.3) + \frac{100}{50}(-0.5) + \frac{100}{10}(0.1)\right) = -(1.5 - 1.0 + 1.0) =$ **−1.5 V**.

**N3. [6 marks]**

a. $I_E = (12 - 0.7)/56\ \text{k}\Omega =$ **201.8 µA**. $I_C \approx I_E/2 =$ **100.9 µA**.
$V_C = 12 - (100.9\ \mu\text{A})(47\ \text{k}\Omega) =$ **7.26 V**.

b. $r_e = 26\ \text{mV}/100.9\ \mu\text{A} =$ **257.7 Ω** (per transistor current).

c. Single-ended: $A_v = R_C/(2r_e) = 47000/515.4 =$ **91.2**. Double-ended:
$A_d = R_C/r_e =$ **182.4**.

**N4. [6 marks]**

a. $P_o = (22 - 2)(0.290 - 0.010)/8 = (20)(0.28)/8 =$ **0.70 W**.

b. $P_i = 12(0.15) =$ **1.80 W**; $\eta = 0.70/1.80 =$ **38.9%** (below 50%).

c. $P_Q = 1.80 - 0.70 =$ **1.10 W**.

d. $R'_L = \Delta V_{CE}/\Delta I_C = 20/0.28 =$ **71.4 Ω**; $a = \sqrt{71.4/8} =$ **2.99 ≈ 3 : 1**.

**N5. [8 marks, chained]**

a. $I_p = 16/16 = 1$ A. $P_i = 2(20)(1)/\pi =$ **12.73 W**. $P_o = 16^2/(2 \cdot 16) =$ **8.0 W**.
$\eta = 8/12.73 =$ **62.8%** (check: $\frac{\pi}{4} \cdot \frac{16}{20} = 0.628$). $P_{2Q} = 4.73$ W, so **2.37 W per transistor**.

b. $P_o(\max) = 20^2/(2 \cdot 16) =$ **12.5 W**.

c. $P_{2Q}(\max) = 2(20)^2/(\pi^2 \cdot 16) =$ **5.07 W** total (2.53 W each), at $V_L(p) = 2V_{CC}/\pi =$ **12.73 V**.

d. Transistor dissipation peaks at about 64% of full swing, where it is higher than at full output, so the heatsinks must handle that worst case.

## 6. Quick recall check

**Question**

1. $V_d = 2$ mV, $V_c = 2$ mV.
2. $10^5$.
3. $1 + 9 = 10$.
4. 25%, 50%, 78.5%.
5. $a = \sqrt{1000/10} = 10:1$.